

Ecology Letters: Topic Modeling and Thematic Alignment Analysis

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1 Introduction

Ecology Letters is a monthly, peer-reviewed journal that publishes research covering all major areas of ecology. Established in 1998, the journal is published by Wiley - an American multinational publishing company, in collaboration with the French National Centre for Scientific Research.

Like in every scientific journal, the *Aims & Scope* of the *Ecology Letters* formally define the thematic boundaries for the papers presented in the journal. The statements declared in the *Aims & Scope* outline topics considered relevant within the journal's domain. However, the thematic composition of published articles may not perfectly correspond to that. Moreover, the thematic patterns in publications tend to group into thematic clusters that shift over time - it seems like a good idea to be able to analyze those changes.

Natural Language Processing is the field which focuses on analysis of large text datasets - and one of the ways in which it can be applied is to identify key themes and trends within them. In this project, *Ecology Letters* is used as a case study, in order to investigate what are the main topics in the published articles, how they align with the journal's stated *Aims & Scope* and how this alignment has evolved over the years. By combining topic modeling, zero-shot classification and semantic similarity measures, the study proposes a framework for analyzing thematic dynamics in scientific journals.

2 Research Questions & Methodology

The primary goal of this study is to develop an NLP-based framework for analyzing and identifying the main thematic topics in published research articles. Using *Ecology Letters* as an example, we will try to apply topic modeling and classification techniques to see what are the dominant research themes present in article abstracts and to examine how these themes evolve over a selected time period.

A second goal is to evaluate the thematic alignment between published research articles and the journal's stated *Aims & Scope*. To achieve this, we will compute semantic similarity scores between article abstracts and the journal's official *Aims & Scope* statement - this will allow us to see the degree of alignment and its changes over time.

The paper addresses the following research questions:

1. What were the dominant topics in articles published in the journal over the last five years?
2. How does the distribution of these topics change over time?
3. To what extent are published articles semantically aligned with the journal's stated *Aims & Scope*?
4. How does the thematic alignment between abstracts and the *Aims & Scope* evolve over time?

To answer these questions, a multi-step NLP framework will be applied. First, all article abstracts are preprocessed using standard text-cleaning procedures. Next, Non-negative Matrix Factorization (NMF) is used to identify key themes in the abstracts. After that, we use zero-shot classification - the modern NLP approach allowing us to double-check and classify the abstracts into specific topics. Finally, semantic similarity measures - cosine similarity in an embedding space - are used to quantify the alignment between each abstract and the journal’s *Aims & Scope* statement. These similarity scores are then analyzed over time to evaluate trends in thematic coherence and consistency, but also to spot outlier papers in the journal - those the least and the most aligned with the *Aims & Scope*.

3 Experimental Results

The collection of abstracts was gathered by using Semantic Scholar API. The idea was to create a dataset of papers - their titles, publication dates and most importantly: the abstracts. At first we got a collection of articles from the period of 2015 - 2025. From what we can see in Table 1, some of the collected articles had missing abstracts - especially in the earlier years. To keep the analysis reliable, we focused on a time frame of 2020 - 2025: the percentage of missing data in this time period is minimal, so it won’t have a major impact on the results. There is one year to be cautious about, 2022, in which there is much less abstracts in total in comparison to other years - it will be taken into consideration while analyzing changes over time.

Year	Total Articles	Missing Abstracts	Fraction Missing
2015	93	75	0.806
2016	125	102	0.816
2017	92	73	0.793
2018	150	70	0.467
2019	43	0	0.000
2020	112	0	0.000
2021	156	0	0.000
2022	76	0	0.000
2023	140	5	0.036
2024	204	8	0.039
2025	233	14	0.060

Table 1: Number of abstracts and missing values per year in the dataset

3.1 Text preprocessing

In order to answer the first research question, and thus find the dominant topics in abstracts published in *Ecology Letters* we need to prepare the data. Our topic modeling relies on traditional NLP techniques that focus on specific keywords, so we developed a preprocessing pipeline to transform the abstracts into a usable format. First, the text was converted to lowercase, any content within square brackets was removed, and the punctuation was stripped. Next, we lemmatized each abstract and removed stopwords - to retain only the meaningful base forms of words. Finally, words were filtered by their part of speech, keeping only nouns and proper nouns to focus on the key concepts and entities in the text. To do this basic text cleaning we used *en_core_web_sm* - a small NLP model from the *spaCy* library.

3.2 Topic modeling

After performing the text preprocessing we can move to the next step - the actual topic modeling. We want to see what are the main thematic clusters visible in our abstracts. To do that, the Non-negative Matrix Factorization will be used. The main idea behind this process is to decompose an original data matrix into two smaller matrices, so that the product of these two would be (approximately) equivalent to the original matrix. Since we will work on matrices, we need to transform our data into this form - we will use the TF-IDF vectorizer for that. After applying that operation on our cleaned abstracts we get the representation of each word that appear in the documents as a value - those values reflect how significant a word is within a specific document compared to the entire corpus.

With over 5,000 unique words across all abstracts, our TF-IDF matrix became quite massive. To make it more manageable, we need to reduce the number of columns by filtering out some of the words. We want to ignore those that are too common, and that is why we will not choose the words that appear in more than 20% of the documents - they won't give us very valuable information about the topics, because they will be too generic. We also don't want the words that are very rare - not helpful for finding some common topics - and to avoid that we choose words that appear in more than 5% of the documents. Together, these settings give us around 140 words - neither too rare nor too common, which will help to extract meaningful patterns from the text.

Having the TF-IDF matrix, we can now do the NMF. In our case the algorithm will approximate the TF-IDF document-term matrix as the product of two smaller ones: the document-topic, and topic-term matrices. The first matrix will allow us to link abstracts to specific topics, while the second one: to define each topic based on clusters of related words.

Since NMF requires us to pre-define the number of topics, we experimented with different settings to see which configuration made the most sense. By analyzing the top words in each cluster, we eventually settled on five distinct groups. Below we can see what are the ten most frequent words in each cluster:

1. network, variation, evolution, host, process, animal, ecology, niche, selection, theory
2. temperature, climate, warming, shift, range, rate, condition, time, variability, year
3. forest, tree, growth, disturbance, density, climate, carbon, productivity, year, survival
4. diversity, biodiversity, relationship, stability, function, richness, scale, loss, conservation, area
5. soil, carbon, biomass, decrease, condition, influence, process, availability, group, experiment

When we look at the clusters we can actually see that they are arranged into some specific thematic groups. The first topic is kind of general, but could be called *evolution*, the second relates mostly to *climate change*, third is about *forests*, the fourth one focuses on *biodiversity*, while the last one mainly on *soils*. This gives us a list of 5 main topics - which we can later analyze. Table 2 shows the percentage distribution of these five topics in the articles in our dataset, containing years from 2020 to 2025.

Topic	Count	Percent (%)
Evolution	376	42.06
Biodiversity	237	26.51
Climate change	135	15.10
Forests	80	8.95
Soils	66	7.38

Table 2: Distribution of dominant topics in the corpus - NMF model results

It looks like the topic of evolution is generally the most popular one, but biodiversity and climate change are also themes significantly present in the articles. Forests and soils are a less popular thematic clusters, but together they account for around 15% of all topics.

Now let us look deeper into the distribution of the topics over time. In the Figure 1 we can see the change in quarters of years 2020 - 2025. The most interesting thing is the fact that in the beginning of our selected time period the evolution topic had almost a 50% share, but it decreased over time. At the end of 2025 the topics are more distributed, with biodiversity and climate change becoming more relevant. When it comes to climate change, we can observe significant fluctuations in earlier years. However, in the last two years, the share of abstracts related to this topic has become bigger and more stable.

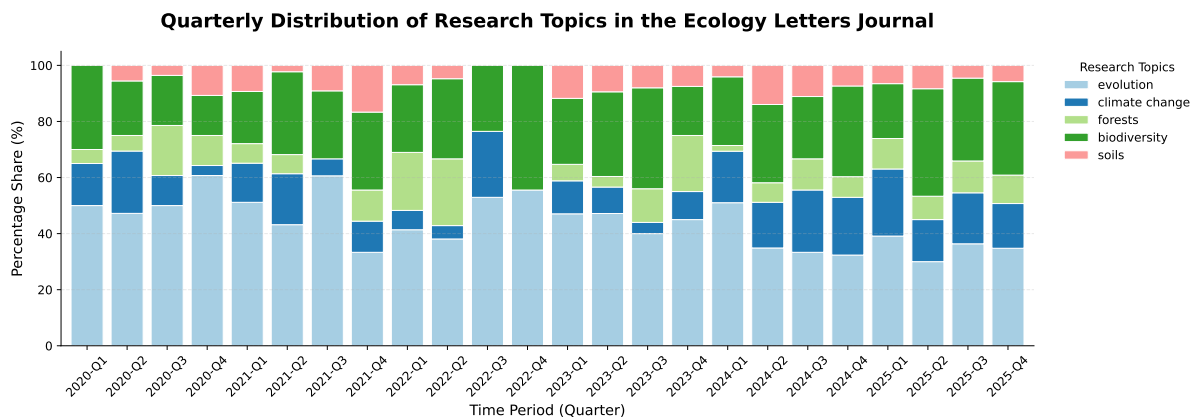


Figure 1: Quarterly distribution of dominant topics in abstracts - NMF model results

It also seems interesting to see which papers were considered to be the top matches within the 5 main topics (list below). When we look at the titles it indeed makes sense that they were attached to these certain themes - the important words for almost all of them actually appear in the titles themselves. The exception is the topic of evolution. It is also a topic with the lowest score. We can explain that by considering this theme as the most general one, not connected to a specific narrow field, so the meaningful words for this topic probably do not have such big scores in comparison to other topics.

1. **Topic: Evolution** (Score: 0.1210)
Title: Genetic correlations and ecological networks shape coevolving mutualisms.
2. **Topic: Biodiversity** (Score: 0.3172)
Title: Species trait diversity sustains multiple dietary nutrients supplied by freshwater fisheries.
3. **Topic: Climate Change** (Score: 0.3494)
Title: Impacts of Weather Anomalies and Climate on Plant Disease.
4. **Topic: Forests** (Score: 0.4136)
Title: Lianas decelerate tropical forest thinning during succession.
5. **Topic: Soils** (Score: 0.4779)
Title: Soil carbon persistence governed by plant input and mineral protection at regional and global scales.

3.3 Zero-shot classification

To create the thematic clusters we used NMF combined with the TF-IDF - the examples of the traditional NLP techniques, which concentrate on statistical frequencies and linear algebra rather than deep learning architectures. Now we can follow another approach, and use the five previously defined topics present in the articles in a zero-shot classification, which is an example of the modern NLP technique. We will use the transformer-based model to get the probabilities of the abstracts relating to one of these 5 topics. The specific model used is [distilbart-mnli-12-3](#) created by Suraj Patil and available on [huggingface.co](#). It's a distilled version of BART, an encoder-decoder model tuned to work in a zero-shot classification. In the traditional NLP techniques the NMF looked at the words inside an abstract, not paying attention to the links between them. In the modern NLP models we have the attention matrices that care of that. That's why we expect a bit different results - which we will later compare.

Topic	Count	Percent (%)
Biodiversity	389	43.56
Evolution	196	21.95
Climate change	155	17.36
Forests	113	12.65
Soils	40	4.48

Table 3: Distribution of dominant topics in the corpus - zero-shot classification

In the Table 3 above we can see that the distribution of topics is much different - in this case the most popular topic is biodiversity, not evolution, like we saw in the NMF method. We expected certain differences because traditional models work differently than modern NLP techniques like zero-shot classification. NMF looked for specific words and how often they appear together - in that case, *evolution* was a broad, general topic, likely based on frequent terms. Modern LLMs, however, understand more from the structure of the abstract, the word order, and the context. This is why the model predicted that the majority of the articles are about biodiversity, rather than evolution in particular.

We identified 382 mismatched abstracts, which accounts for nearly 43% of the total dataset. Interestingly, in 164 of these cases, NMF classified the abstract as *evolution*, while the zero-shot model categorized it as *biodiversity*. This suggests that for modern NLP models, the *biodiversity* cluster was a more precise choice for a general topic. To better understand these discrepancies, we can examine some examples of papers where the models disagreed. In the evolution/biodiversity mismatch category, titles include:

- ”Best be(e) on low fat: linking nutrient perception, regulation and fitness.”
- ”Integrating data mining and transmission theory in the ecology of infectious diseases”
- ”Co-occurrence is not evidence of ecological interactions.”

These topics actually do not align perfectly with any of our predefined thematic clusters, which explains why the models gave different outcomes. However, even in these ambiguous cases, the biodiversity cluster appears to be a more suitable choice than evolution. Of course, the naming of these topics was a subjective decision, and the results might vary if the clusters were labeled differently. Nevertheless, it is fascinating to observe these distinct behaviors across different modeling techniques.

3.4 Measuring similarity with Aims & Scope

One of the research questions the paper tries to answer is measuring the extent to which the published articles semantically align with the journal’s stated Aims & Scope. The official statement regarding thematic boundaries for the papers of *Ecology Letters* can be read [here](#).

Since we have a clearly defined Aims & Scope, we can calculate the similarity scores between that and the abstracts of the papers present in the journal. The first step is to have a vector representation of our data. To do that we will use the embeddings of the encoder-only model downloaded from [huggingface.co](#). In this kind of an exercise we need a model that can give us a vector representation in a multi-dimensional space - encoder models do exactly that. The model we are going to use is [bge-base-en-v1.5](#) - a feature extraction model created by the Beijing Academy of Artificial Intelligence.

We can apply the *feature_extractor* pipeline to extract the embedding of our abstracts and the Aims & Scope. The embedding space has 384 dimensions, and we can locate each one of the documents somewhere in this space. Now, since the units have a vector representation we can finally calculate the similarity score between them. The similarity score will be calculated by using a cosine similarity. It works by calculating the cosine of the angle between the vectors, telling us if they are moving in a similar direction.

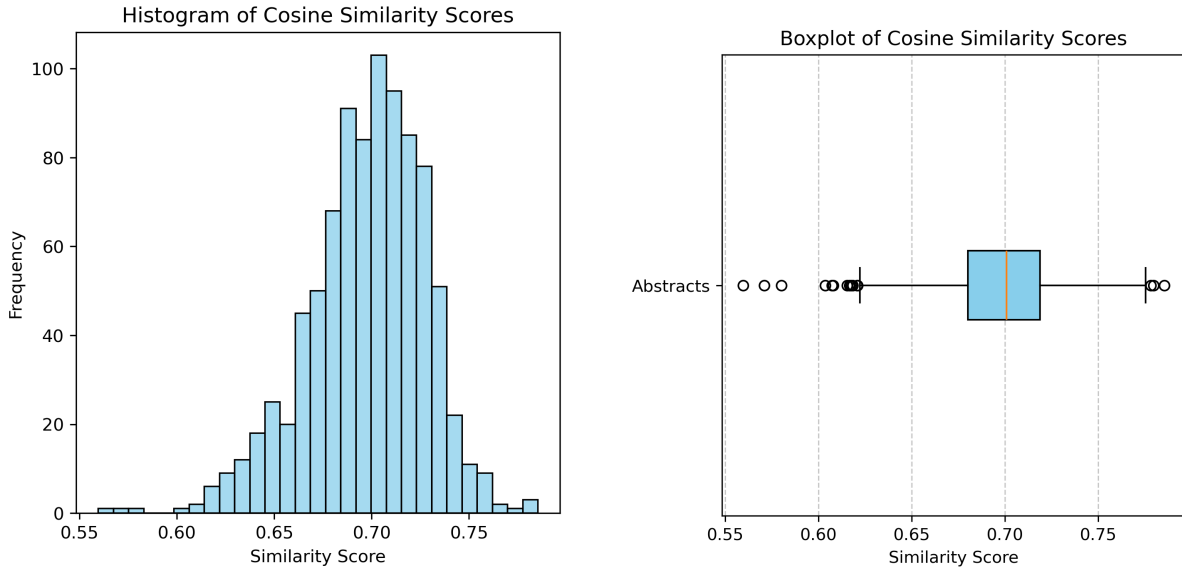


Figure 2: Distribution of the similarity scores between abstracts and Aims & Scope

In the Figure 2 we can see the distribution of the cosine similarity scores between the journal’s Aims & Scope and our abstracts. From the histogram we learn that the majority of scores is around 0.7 and that the distribution is quite dense. The value itself doesn’t tell us a lot, so to understand more about this score we can look for the outliers and compare the scores between different types of abstracts.

Boxplot can help us better visualize the results even better. We can see that the median is a bit over 0.7, and the IQR range is between 0.68 and 0.72, which proves a strong thematic coherence. What is more, we can see a couple of outliers - on both sides, so we have some abstracts that are not very well aligned with the Aims & Scope, but also some of them that have a very high score compared to the rest. Let’s see which are the ones with the lowest and highest score.

Rank	Score	Title
Lowest	0.5596	Issue Information.
2nd Lowest	0.5710	Contemporary selection on MHC genes in a free-living ruminant population.
3rd Lowest	0.5803	Paternal age negatively affects sperm production of the progeny.

Table 4: Articles with the lowest cosine similarity to Aims & Scope

As it turns out, we had one abstract that wasn’t a null value, but is not a real abstract (the abstract was actually a *”No abstract is available for this article”*). No wonder that we got this as an outlier, with the worst thematic alignment. The 2nd and 3rd papers with lowest scores are, from what we can understand from the titles, more about the biology and genetics rather than the ecological issues. That is why they were classified so low. Now, to compare, let us see what was the paper with the highest alignment score:

Highest Similarity Score: 0.7854

Title: Opportunities to advance the synthesis of ecology and evolution.

It makes perfect sense that this paper is the one with the highest thematic alignment - it talks about the evolutionary ecology (what was mentioned in the keywords of the Aims & Scope), and presents some new ideas in the field.

How can we confirm that these similarity scores are truly high? One effective method is to compare our results with abstracts from completely different fields to provide a necessary contrast. We chose one abstract from the field of Quantum Physics, and another one from the field of Social Sciences. After

computing the cosine similarity score between these abstracts and Ecology Letters’ Aims & Scope we got the results: 0.5192 and 0.4995. There is a significant difference between these scores, and the ones coming from the papers included in the journal.

The last question we want to answer is about the thematic drift in the papers from the journal, in the span of 5 years (2020 - 2025). When we look at the Figure 3 showing the quarterly similarities of papers’ abstracts vs Aims & Scope over time we don’t really see a significant trend - except for the peak in the 4th quarter of 2022. Although the average score for this period is higher, the 95% confidence intervals are nearly twice as wide as in other years, indicating significantly greater variability. This is likely due to the limited data available for 2022, particularly in the fourth quarter, which recorded the lowest number of abstracts. Excluding this outlier, we observe the confidence intervals narrowing over time as the sample size increases. In general, we can say that while the average thematic alignment remained relatively constant, the results became more stable with fewer fluctuations in the later years of the study.

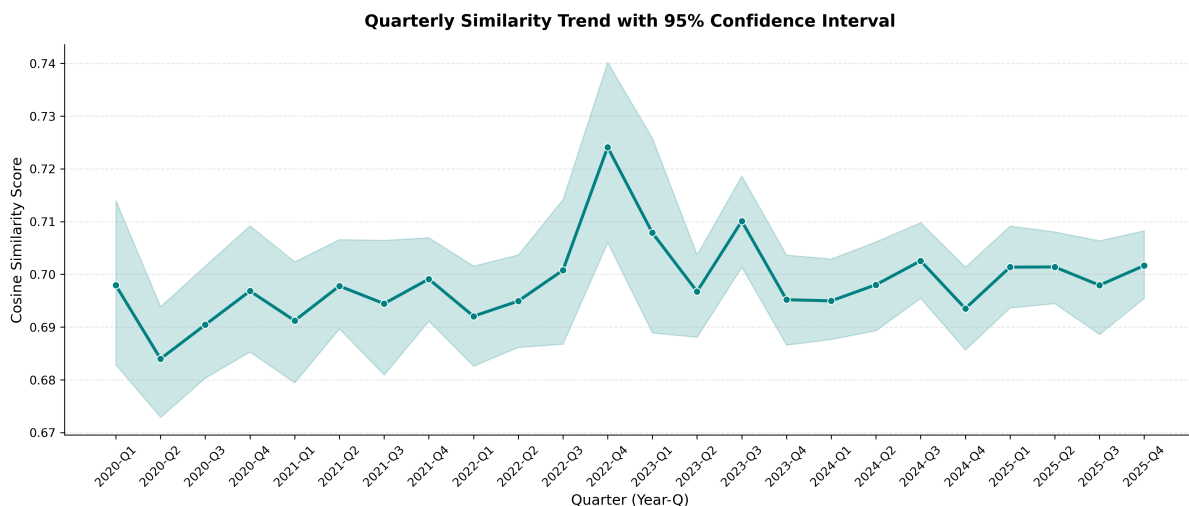


Figure 3: Cosine similarity scores over time

4 Concluding Remarks

In summary, we conducted an analysis of topic modeling and thematic alignment using *Ecology Letters* as an example. We found 5 main topic clusters in the articles: biodiversity, evolution, climate change, forests, and soils. The distribution of these topics varied depending on the quarter and year; in later years, the topics became more diversified, with climate change specifically becoming more relevant. We observed that the differences between traditional and modern NLP techniques are significant. They focus on different aspects, and in this case, the modern NLP models performed better.

Next, we analyzed the alignment scores between the abstracts and the journal’s Aims & Scope. The outliers we identified did not correspond well to the journal’s statements. Comparing abstracts to these outliers, but also to articles from different fields helped us confirm that the alignment for papers in *Ecology Letters* was significantly higher. The analysis of similarities over time showed that the scores are steady. We do not see a divergence between the topics in the papers and the Aims & Scope statements - in fact, there is even a slight positive trend.

After creating this NLP pipeline for analyzing scientific journals, it would be a good idea to conduct the same analysis for different journals and compare the results. Due to computational constraints, the models used in this paper were small - it would be useful to see how the results might change using more advanced tools.

AI Usage Disclaimer

Artificial Intelligence (OpenAI's ChatGPT 5 and Google Gemini) was used during the development of this project to assist with Python coding and to enhance the linguistic quality of the written report. While the AI provided support in structuring functions and refining the academic tone, the final content, logic, and data interpretation were critically reviewed and finalized by the author and they take full academic responsibility and integrity for this work.